# ADHR-CDNet: Attentive Differential High-Resolution Change Detection Network for Remote Sensing Images

Xiuwei Zhang, Mu Tian, Yinghui Xing, *Member, IEEE*, Yuanzeng Yue, Yanping Li, Hanlin Yin, Runliang Xia, Jin Jin, and Yanning Zhang, *Senior Member, IEEE*

***Abstract*— With the development of deep learning, change detection technology has gained great progress. However, how to effectively extract multiscale substantive changed features and accurately detect small changed objects as well as the accurate details is still a challenge. To solve the problem, we propose attentive differential high-resolution change detection network (ADHR-CDNet) for remote sensing images. In ADHR-CDNet, a novel high-resolution backbone with a differential pyramid module (DPM) is proposed to extract multilevel and multiscale substantive changed features. The backbone structure with four interconnected subnetwork branches of different resolution is helpful to extract multilevel and multiscale features. DPM is capable of distinguishing between substantive changes and pseudochanges induced by illumination, shadow, seasonal variation, and so on. Then, a novel multiscale spatial feature attention module (MSSAM) is presented to effectively fuse the spatial detail information of different scale features produced by our backbone to generate finer prediction. We conduct quantitative and qualitative experiments on three public change detection datasets: the Lebedev, LEVIR building change detection dataset (LEVIR-CD), and Wuhan University (WHU) Building datasets. The proposed ADHR-CDNet reaches F1-score of 97.2% (improved 3.1%) on the Lebedev dataset, 91.4% (improved 1.6%) on the LEVIR-CD dataset, and 90.9% (improved 1.2%) on the WHU Building dataset. The experimental results demonstrate that our method performs much better than the state-of-the-art (SOTA) methods. The visualization comparison results show that our method can effectively detect small changed objects and significantly improve the details of detected changed objects. Our code is available at https://github.com/w-here/ASGO-113lab/tree/main/ADHR-CDNet.**

***Index Terms*— Attention module, change detection, deep learning, differential information, remote sensing.**

## I. Introduction

CHANGE detection is one of the most important tasks in the field of Earth observation image analysis. The aim of it is to identify the changed information of ground objects by analyzing and comparing the remote sensing image pairs taken at different times in the same area. This technology plays an important role in the fields of land use, urban coverage, disaster monitoring, civil infrastructure, environmental monitoring, and so on [1], [2], [3], [4], [5], [6], [7], [8].

Now, with the development of remote sensing big data and deep learning, change detection field has reached the third climax (after 2010). The change detection methods based on deep learning have gained great progress and can be divided into three categories according to whether they use label information: supervised methods [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], unsupervised methods [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], and semisupervised methods [44], [45], [46]. The supervised one is the primary focus of this article.

Recent supervised change detection methods can be divided into two groups. The first group is based on metric learning [10], [11], [12], [13]. They calculate the distance between two bitemporal feature maps and then use the similarity metric to obtain the change map. The bitemporal feature maps are usually produced by deep convolution neural network, such as Residual Network (ResNet) [47] and visual geometry group (VGG) [48]. Shi et al. [13] proposed a deeply supervised attention metric-based network (DSAMNet) and used the convolution block attention module to provide more discriminative features. The second group is based on the semantic segmentation network like fully convolutional networks (FCNs) [49]. CDNet [15] was a typical and fundamental FCN-based network, that which consisted of

Manuscript received 6 May 2022; revised 31 August 2022 and 17 October 2022; accepted 7 November 2022. Date of publication 10 November 2022; date of current version 21 November 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61971356, Grant U19B2037, and Grant 62201467; in part by the National Natural Science Foundation of Shaanxi Province under Grant 2021KWZ-03 and Grant 2022JQ-686; in part by the Major Scientific and Technological Special Project of Henan Province under Grant 201400211000; and in part by the Science Foundation for Excellent Young Scholars of Henan Province under Grant 212300410059. *(Corresponding author: Xiuwei Zhang.)*

Xiuwei Zhang, Mu Tian, Yinghui Xing, Yuanzeng Yue, Yanping Li, Hanlin Yin, and Yanning Zhang are with the National Engineering Laboratory for Integrated Aerospace-Ground-Ocean Big Data Application Technology, Xi'an 710072, China, also with the Shaanxi Provincial Key Laboratory of Speech and Image Information Processing, Xi'an 710072, China, and also with the School of Computer Science and Technology, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: xwzhang@nwpu.edu.cn; 291775319@qq.com; xyh_7491@163.com; 651570714@qq.com; yanpinglics@gmail.com; iverlon1987@126.com; ynzhang@nwpu.edu.cn).

Runliang Xia is with the Yellow River Institute of Hydraulic Research, Zhengzhou 450003, China (e-mail: xiarunliang@hky.yrcc.gov.cn).

Jin Jin is with the School of Computer Science and Technology, Northwestern Polytechnical University, Xi'an 710072, China, and also with the Yellow River Institute of Hydraulic Research, Zhengzhou 450003, China (e-mail: jinjin@hky.yrcc.gov.cn).

Digital Object Identifier 10.1109/TGRS.2022.3221492

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

a contraction network and an expansion network. Daudt et al. [16] proposed three change detection networks based on FCN, including fully convolutional-early fusion (FC-EF), FC-Siam-conc, and FC-Siam-diff. Liu et al. [17] proposed a deep Siamese convolutional network model with dual-task constraints (DTCDSCNs). Chen et al. [20] proposed a double attention full convolution Siamese network (DASNet) for change detection of high-resolution images. Chen and Shi [21] proposed spatial–temporal attention-based network (STANet). In addition, there are some variants of FCN, such as U-shaped network (U-Net) and U-Net++. Peng et al. [22] adopted an effective deep supervised strategy for remote sensing image change detection (Improved UNet++). Zhang et al. [23] enhanced UNet++ with differential features to improve the change detection performance (DifUNet++). Later, a high-resolution network (HRNet) [50] was proposed, which was initially used for human pose estimation and subsequently achieved good results in the field of change detection. Cao et al. [24] proposed a Siamese high-resolution network (Si-HRNet) to improve the effect of small regions' change detection. Pan et al. [25] adopted a high-resolution network as backbone and proposed a new temporal reliable change detection (TRCD) algorithm. Besides, with the development of multimodel image semantic segmentation [51], [52], [53], many scholars began to pay attention to multimodel change detection [36], [41], [54].

The recent state-of-the-art (SOTA) methods based on semantic segmentation model structures have continuously improved the performance of remote sensing image change detection. However, it remains a challenge to accurately detect small changed objects and the details. Two main problems should be solved at the same time: 1) the scale diversity of changed objects and 2) substantive change information robust extraction, unaffected by significant differences in visual appearances between two temporal images caused by the influence of illumination, shadow, seasonal variation, and so on [10], [21], [32], [55], [56]. Most recent methods adopted the backbone with multiscale feature extraction ability [such as Unet++, HRNet, and atrous spatial pyramid pooling (ASPP)] and difference strategy to achieve multiscale change detection. Therefore, how to effectively extract multiscale context semantic features and multiscale substantive difference information and further aggregate them to accurately detect small changed objects and the details is a key to further improve change detection precision.

To solve the above problem, in this article, we propose attentive differential high-resolution change detection network (ADHR-CDNet) for remote sensing images. The main contributions are summarized as follows.

1) We design a novel high-resolution backbone with a differential pyramid module (DPM) for change detection. DPM is conductive to distinguish between substantive changes and pseudochanges. The backbone is made up four interconnected subnetwork branches of varying resolutions, one of which has the same resolution as the input image. The multiscale explicit differential features produced by DPM are integrated with the four interconnected branches. Therefore, our backbone can extract multilevel and multiscale substantively changed features while retaining high-resolution finer features for encoding small objects.
2) Before prediction, we design a multiscale spatial feature attention module (MSSAM) inspired by [57] to effectively fuse the spatial detail information of different scale feature groups produced by our backbone. Different from the existing attention mechanisms, MSSAM can automatically learn a weight map for the feature group of each scale. The change detection accuracy is further improved using MSSAM.
3) The proposed ADHR-CDNet model significantly improves the performance of change detection. We conduct quantitative and qualitative experiments on three public change detection datasets: the Lebedev, LEVIR building change detection dataset (LEVIR-CD), and Wuhan University (WHU) Building datasets. ADHR-CDNet reaches an F1-score of 97.2% on the Lebedev dataset, 91.4% on the LEVIR-CD dataset, and 90.9% on the WHU Building dataset. The quantitative experimental results demonstrate that our method outperforms the SOTA methods on these three datasets. The visualization results show that our method can detect both large-scale and very small-scale changed objects and can significantly improve the details of detected changed objects.

The remainder of this article is organized as follows. Section II is an overview of the related work. Section III describes the architecture of ADHR-CDNet and the details of four components including backbone, DPM, MSSAM, and loss function. Section IV presents the experiment configuration and analyzes the experimental results. Finally, Section V concludes our findings and future work.

## II. Related Work

In recent years, many scholars have conducted extensive researches on change detection based on deep learning.

In the beginning, since the lack of available large-scale change detection dataset, deep belief networks [28], [42] were adopted as a discriminant model to determine whether an area has changed or remain unchanged. Zhang et al. [28] created the changed map by combining feature changed analysis and deep belief networks. Subsequently, with the advent of large-scale change detection datasets, many methods used convolutional neural networks (CNNs) to extract abstract and robust features to improve the change detection performance. Liu et al. [9] proposed a dual-channel CNN model to detect changes from bitemporal synthetic aperture radar (SAR) images. Zhan et al. [11] built a Siamese CNN network for change detection with contrastive loss. Zhang et al. [12] used a Siamese CNN network for change detection as well, but instead of contrastive loss, they used an improved triplet loss. Bai et al. [14] proposed edge-guided recurrent convolutional neural network (EGRCNN), which combined identification information and edge information to build priorities in a framework to improve the change detection results, especially to generate more accurate building boundaries.

While the classical CNN uses a full connection layer after convolution layers to create a fixed length feature vector for classification, it is unable to perform pixel-level prediction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

directly. The FCN [49] was presented as a solution to this problem. FCN replaces the full connection layers with the deconvolution layers, which upsamples the feature maps to match the input image's size. As a result, it can produce pixel-level prediction maps directly. FCN is frequently used for semantic segmentation. Its design concept is also suitable for change detection, which can be regarded as a binary semantic segmentation task: changed area and nonchanged area.

Therefore, FCN has become the backbone of many recent change detection methods. CDNet [15] was a typical and fundamental FCN-based network that consisted of contraction network and expansion network. These two parts were used to extract features and locate the changed area. Daudt et al. [16] proposed three FCN-based change detection networks: FC-EF, FC-Siam-conc, and FC-Siam-diff. FC-EF directly used the concatenation of two input images as the input to FCN. FC-Siam-conc had two Siamese encoders for two input images, respectively, and one decoder. Skip connections were designed to connect the corresponding layers from the two encoders and the decoder. FC-Siam-diff was similar to FC-Siam-conc, but it took into account the absolute difference between the two encoders' feature maps. Liu et al. [17] proposed a dual-task constrained deep Siamese convolutional network model (DTCDSCN) that could perform pixel-level building change detection task and object-level building segmentation task simultaneously. Ding et al. [18] proposed a deeply supervised attention-guided network (DSA-Net), which introduced a spatial attention-mechanism-guided cross-layer addition and skip-connection (CLA-Con-SAM) module to aggregate multilevel contextual information. Wang et al. [19] proposed an attention-based deeply supervised network (ADS-Net). To highlight the characteristics of change, an adaptive attention mechanism was proposed to capture the relationship of different scale changes by combining features of space and channel.

There were also some methods that focused on designing attention module for change detection, with the goal of filtering or reducing the pseudochanges caused by interference factors. Chen et al. [20] proposed a DASNet for high-resolution image change detection. Its dual-attention module contained a position attention module (PAM) and channel attention module (CAM), which significantly improved the performance of change detection. Chen and Shi [21] proposed STANet, which used a self-attention module based on space–time dependence to calculate the attention weight of any two pixels at different times and positions. This attention module improved the relationship between pixels in an image that were far apart.

Subsequently, U-Net was proposed by Ronneberger [58], which greatly increased the performance of semantic segmentation, especially improving the detail of the segmentation results. It benefited from the structure of skipping connection between the corresponding layers of the encoder and the decoder. U-Net was also used to solve the change detection problem. Lei et al. [27] designed difference enhancement and spatial–spectral nonlocal (DESSN) network for change detection in very-high-resolution (VHR) images. Zhou et al. [59] proposed U-Net++. Compared with U-Net, it contained more intensive skip connections and used the deeply supervised multiside output fusion method. U-Net++ has the potential to further increase the accuracy of semantic segmentation, especially for high-resolution images. It was also used in the field of change detection. Peng et al. [22] adopted an effective deep supervised strategy for remote sensing image change detection. Zhang et al. [26] designed a deeply supervised image fusion network (INF) and used a deep supervision strategy to solve the disappearance of gradients during feature fusion. DifUNet++ [23] added differential features to U-Net++ to enhance change detection performance. DifUNet++Dupsampling (DU) [23] applied a learnable upsampling strategy to DifUNet++ for more accurate detection. The transformer was then gradually applied to the field of computer vision, with promising results.

In recent years, HRNet was proposed for human pose estimation. It can maintain high-resolution representations at all times, not just by recovering high-resolution representations from low-resolution representations. In this way, the effect of gesture recognition is significantly improved. Later, in the field of change detection, some scholars began to adopt HRNet as their backbone. Cao et al. [24] proposed an Si-HRNet to improve the effect of small region change detection and proposed a new hybrid loss function to further improve the performance. Pan et al. [25] adopted HRNet as the backbone and proposed a new TRCD algorithm. Chen et al. [29] applied transformer to change detection field and proposed bitemporal image transformer (Bit).

## III. Methodology

The proposed ADHR-CDNet for remote sensing image change detection is elaborated in this section. As shown in Fig. 1, ADHR-CDNet contains three parts: the DPM, the modified high-resolution network (MHRNet), and the MSSAM, which are noted as Part A, Part B, and Part C, respectively. Among them, Part A and Part B are combined and served as our backbone.

The explicit differential information of the input image pair is helpful for extracting and encoding changed information. Therefore, DPM is designed to extract and encode multiscale differential feature maps. The detail of DPM (Part A) is described in Section III-A. High-resolution features are beneficial for extracting small changed area, so we design our backbone based on HRNet [50]. Specifically, we modify the architecture of HRNet to take full advantage of multiscale differential feature produced by DPM and adapt it to our fusion model MSSAM. The detail of MHRNet (Part B) is illustrated in Section III-B. To sufficiently make use of different scale features extracted by the backbone to generate a precise prediction, an effective feature fusion strategy is required. In our architecture, MSSAM (Part C) is designed to fuse multiscale features, which is presented in Section III-C. Finally, Section III-D introduces the loss function.

### A. Differential Pyramid Module (DPM)

The explicit differential information is beneficial for extracting and encoding the changed features. Therefore, we design the DPM that can extract multiscale differential features from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

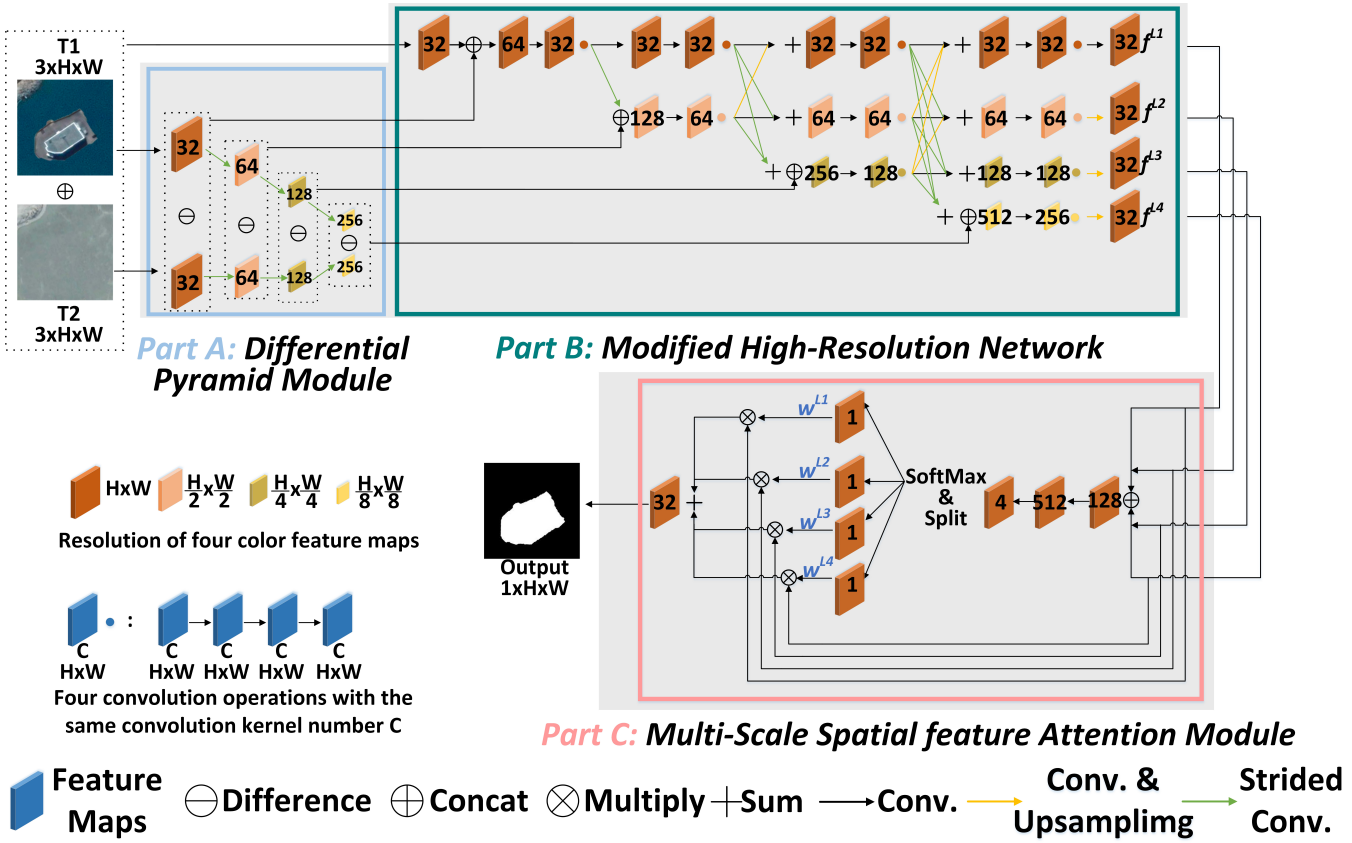



Fig. 1. Architecture of the proposed ADHR-CDNet.

two input images. In addition, the extracted multiscale differential feature maps, totally four levels, are used as the partial input to four subnetworks of our backbone, respectively.

As shown in Part A of Fig. 1, the two input images are noted as T1 and T2, which were taken at different times in the same area. The input image pair is fed into DPM. For each input image, it goes through four convolution layers. The first layer adopts $3 \times 3$ convolution and the others adopt $3 \times 3$ convolution with stride 2. The resolutions of the feature maps produced by these four layers are one time, one-half, one-quarter, and one-eighth of the input image size, respectively. Their number of channels is 32, 64, 128, and 256, respectively. Then, the absolute difference is applied to calculate the differential information between the corresponding feature maps of these four layers for two input images. As a result, DPM generates four levels of differential feature maps with one time, one-half, one-quarter, and one-eighth of the input image size.

### B. MHRNet

As shown in Part B of Fig. 1, MHRNet is modified from HRNet to adapt to multiscale differential features' input and our fusion model MSSAM. From top to bottom, it has four parallel subnetwork branches labeled L1–L4. L1 is the top branch with one convolution layer and four convolution blocks. Each convolution block includes five convolution layers, the last four of which are illustrated in Fig. 1 by a feature map and a little circle to simplify the architecture figure. The spatial sizes of all the feature maps in L1 are the same as that of the input, and therefore, the backbone can keep high-resolution features beneficial for small changed object detection. The structures of the other three branches are similar to L1. Specifically, L2, L3, and L4 have three, two, and one convolution block(s), respectively. The feature maps for each subnetwork have the same spatial size. The spatial sizes of the feature maps of L2–L4 are 1/2, 1/4, and 1/8 of the input image size, respectively. The kernel size of the convolution operations in L1–L4 is $3 \times 3$ unless otherwise stated. The number of channels of all the feature maps is marked in Fig. 1.

These four subnetworks are not independent but interconnected. In Fig. 1, the black, yellow, and green arrows represent a residual unit, convolution and upsampling operation, and a strided convolution, respectively. They are adopted to implement information exchange among different subnetwork branches, which is similar to HRNet. Note that the input to the first convolution block of each subnetwork contains two parts: one part from DPM and the other from the outputs of the topper branch(es) or the output of the previous convolution (only for L1). If the second part is from more than one branch, these feature maps are first summarized, and then concatenated with the feature maps from DPM; otherwise, the feature maps from two parts are concatenated. Finally, four feature maps produced by four branches are upsampled to generate four

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

output feature maps, namely, $f^{L1}$–$f^{L4}$, which have the same resolution and the same number of channels, as shown at the rightmost end of part B. These four groups of upsampled feature maps are input to MSSAM.

### C. Multiscale Spatial Feature Attention Module (MSSAM)

The coexistence of multiscale targets is a common phenomenon in remote sensing images. An effective feature fusion strategy is necessary to sufficiently take advantages of varied scale features produced by the backbone. Therefore, we design our MSSAM inspired by [57]. As shown in Part C of Fig. 1, MSSAM differs from the existing attentions such as position attention [60], channel attention [60], and residual attention (RA) [61], and it is a kind of multiscale positional attention, designed to learn the weight maps for different scale feature groups. $f^{L1}$–$f^{L4}$ obtained from Part A are used as the input to MSSAM. First, $f^{L1}$–$f^{L4}$ are concatenated; then, two convolution operations, which have 512 filters with the kernel size $3 \times 3$ and four filters with the kernel size $1 \times 1$, are applied to the concatenated result sequentially and generate four feature maps noted as $h^s$ ($s \in S = \{L1, L2, L3, L4\}$). After that, $h^s$ are normalized using a softmax operation to generate four weight maps $w^s$ ($s \in S$). The calculation formula of the weight map is given as

$$w_i^s = \frac{\exp\left(h_i^s\right)}{\sum_{t=1}^{S} \exp\left(h_i^t\right)} \tag{1}$$

where $i$ ranges over all the spatial positions. Finally, the weight map $w^s$ is multiplied with the feature maps $f^s$, and then the weighted feature maps are summed up to produce the fusion result $g$. The calculation procedure is depicted by (2)

$$g_{i,c} = \sum_{s=1,c=1}^{S,32} w_i^s \cdot f_{i,c}^s \tag{2}$$

where $g_{i,c}$ is the weighted sum score at position $i$ of channel c at all the scales.

Fig. 2 shows the visualization of MSSAM on four scenes. In Fig. 2, (a) and (b) are the bitemporal remote sensing images of the same area, respectively, (c) are their labels, and (d) and (e) are the heat maps of the network with and without MSSAM respectively. We can see that our network gives more weight to the pixels of the substantive changed area and ignores the pixels of the pseudochanged area caused by seasonal change. Moreover, the details of the changed objects, especially the edges and outlines, are clearer using MSSAM.

### D. Loss Function

Binary cross-entropy loss is widely used in the binary classification tasks. The change detection task in our article can be regarded as a pixel-level binary classification task (the changed pixel and the unchanged pixel). So, we select binary cross-entropy loss as our loss function, which is described as follows:

$$L_{\text{bce}}(y, x) = -\frac{1}{N} \sum_i \{y_i \times \log(x_i) + (1 - y_i) \times \log(1 - x_i)\} \tag{3}$$

where $N$ is the number of pixels in an image patch, $y$ is a change label map, $y_i \in \{0, 1\}$ represents the value of position $i$ in $y$ (here, 1 and 0 mean the pixel has changed and unchanged, respectively), $x$ represents the predicted change map of the network model, $x_i \in [0, 1]$ is the value of position $i$ in $x$, and the value of $x_i$ indicates the probability of the pixels $i$ detected as a changed point.

## IV. Experiments

### A. Datasets

To verify the effectiveness of the proposed ADHR-CDNet, we adopt three public remote sensing image change detection datasets including the Lebedev dataset [32], the LEVIR-CD dataset [21], and the WHU Building dataset [56].

The Lebedev dataset has seven pairs of seasonal image with a size of $4725 \times 2200$ pixels and four pairs of images with a size of $1900 \times 1000$ pixels. The resolution of these image pairs varies greatly, ranging from 3 to 100 cm per pixel. The seasonal difference between image pairs is particularly significant. The changed area in this dataset covers all kinds of changed objects in the captured images, such as cars, buildings, roads, and land. Therefore, this dataset is very challenging. When making the label of the changed area of the dataset, the differences caused by factors, such as season, light, and brightness, are ignored, and only substantive changes are retained. The dataset is randomly cropped to 16 000 pairs of image patches with a size of $256 \times 256$ pixels [32]. Among them, 10 000, 3000, and 3000 pairs of image patches are used for training, verification, and testing, respectively, so their ratio is 10:3:3. Note that all 16 000 pairs of image patches contain the changed area.

The LEVIR-CD dataset focuses on building change detection and contains 637 high-resolution (0.5 m/pixel) remote sensing image pairs with a size of $1024 \times 1024$ pixels. These bitemporal images were captured from 20 different locations in various Texas cities in the US, including Austin, Lakeway, Bee Cave, Buda, Kyle, Manor, Dripping Springs, and so on. These images of different regions were taken from 2002 to 2018 at different times. It also considers seasonal changes and the changes caused by light, shadows, etc. The dataset only focus on the changes in buildings, and construction of roads on land is not included in the changes. The dataset is randomly divided into three parts according to the ratio of 7:1:2 by [21] for training, verification, and testing, respectively. Specifically, their numbers of image pairs are 445, 64, and 128, respectively. To better train the network, the original large-scale remote sensing images are cut into image patch pairs with a size of $256 \times 256$ pixels. Finally, 7120, 1024, and 2048 pairs of image patches are used for training, verification, and testing, respectively.

The WHU Building dataset is mainly used for building extraction and building change detection [56]. It contains a pair of images of the same area before and after an earthquake disaster of the magnitude 6.3 ($15\,354 \times 32\,507$ pixels), their corresponding building detection label maps, and a corresponding black-and-white binary building change map. The image in the first phase was collected in April 2012 covering

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

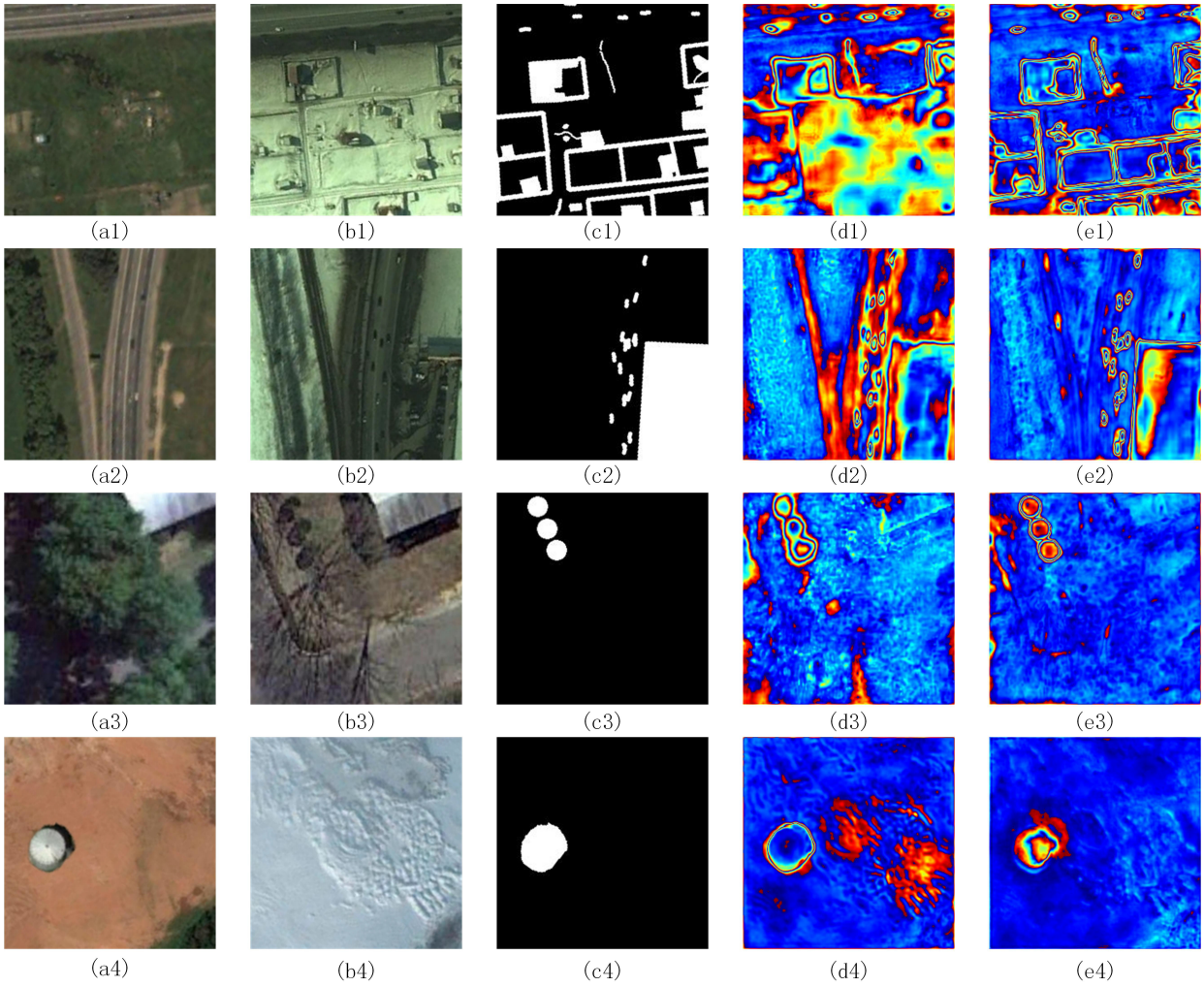



Fig. 2. Comparison of heat maps of MSSAM. (a) T1 image. (b) T2 image. (c) Ground truth. (d) Heat maps of the network without MSSAM. (e) Heat maps of the network with MSSAM.

Christchurch, New Zealand, with a resolution of 1.6 m (12 796 buildings within 20.5 square kilometers). The image in the second phase was taken in 2016 (the number of buildings in the same area became 16 077). This dataset only focuses on the changes in large and sparse buildings. The emergence or disappearance of cars and other nonbuilding changes are ignored. The WHU Building dataset is cropped into 7620 pairs of small image patches with a size of 256 × 256 pixels through a sliding window with stride 256, of which 6096, 762, and the remaining 762 image pairs are used for training, validation, and testing, respectively, so their ratio is 8:1:1 [29].

To avoid overfitting during network training, we augmented and expanded the training set of the adopted three datasets used in the experiments. Each image pair is rotated by 90°, 180°, and 270° and also horizontally and vertically flipped. After augmentation, the amount of training data becomes six times of the original.

### B. Implementation Details

The proposed network is implemented by PyTorch and runs on a workstation with Intel(R) Xeon(R) CPU E5-2678 (2.50 GHz, 12 cores) and a graphics processing unit (GPU) of NVIDIA 2080 Ti. The optimization algorithm uses stochastic gradient descent (SGD). The base learning rate lr is initially set to 0.001 and decays as $\mathrm{lr} = \mathrm{lr} * 0.6^{\mathrm{epoch}/2}$, where epoch represents the number of times that the current network has been trained on all the training samples. The batchsize is set to 4, and the training time is set to 100 epochs. The patch size of the three datasets is 256 × 256 × 3.

### C. Evaluation Criteria

In the field of change detection, there are three commonly used evaluation criteria, named Precision, Recall, and F1-score, as shown in (4)–(6), respectively. In this article, we also adopt these three criteria to measure the performance of the proposed network. Precision represents the positive predictive rate. The higher the Precision, the fewer false alarms. Recall represents the true positive (TP) rate. The larger the Recall, the less the number of missed detection. F1-score is an overall measure combining Precision and Recall; that is, a higher F1-score represents lower false positive (FP) and lower false negative (FN)

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{4}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

$$F_1 = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}. \quad (6)$$

Among them, TP denotes the number of correctly classified positive pixels in the predicted change map, FP denotes the number of incorrectly predicted positive pixels in the change map, and FN denotes the number of incorrectly predicted negative pixels in the change map.

### D. Comparison Methods

The proposed model is compared with 11 SOTA methods, including CDNet [15], FC-EF [16], FC-Siam-conc [16], FC-Siam-diff [16], DTCDSCN [17], DASNet [20], STANet [21], DifUNet++ [23], DifUNet++DU [23], Bit [29], DSA-Net [18], ADS-Net [19], DSAMNet [13], EGRCNN [14], Si-HRNet [24], and TRCD [25]. Among them, CDNet is a typical and basic FCN-based change detection network. FC-EF, FC-Siam-conc, and FC-Siam-diff are also FCN-based change detection networks, but with different encoder structures. DTCDSCN is an FCN-based Siamese change detection network. DASNet and STANet, respectively, propose a dual-attention module and a spatial–temporal attention module for the FCN-based change detection network. DifUNet++ and DifUNet++DU are based on U-Net++. Bit applies a transformer to the field of change detection. DSA-Net designs a dual-branch feature extraction network and the extracted features are transmitted to the difference detection network to detect changes. ADS-Net uses a pair of Siamese FCN networks to extract features and designs an attention mechanism to fuse them. DSAMNet uses the convolution block attention module to provide more discriminative features. EGRCNN combines identification information and edge information to build priorities in a framework to improve the change detection results. Si-HRNet designs a Siamese high-resolution network to improve the effect of small regions' change detection. TRCD adopts high-resolution network as backbone and proposes a new TRCD algorithm. All the comparison methods except for DSA-Net adopt the same dataset division way described in Section IV-A. DSA-Net crops the images into 128 × 128 pixels image patches. At the same time, we design a lightweight ADHR-CDNet [namely, ADHR-CDNet(S)], which is different from ADHR-CDNet in that the input image undergoes two extra strided convolutions with a stride of 2 before the first convolution block of DPM, so that the resolution becomes a quarter of the original.

### E. Performance Comparison

*1) On the Lebedev Dataset:*

*a) Quantitative evaluation:* Table I shows the quantitative comparison results with the SOTA methods on the Lebedev dataset. The best performance is highlighted in red, and the second best is highlighted in blue, as is the case in subsequent tables. It can be seen from Table I that ADHR-CDNet outperforms all the comparison methods and achieves the highest F1-score (97.2%). Compared with the second-best one (DASNet), ADHR-CDNet increases the accuracy of F1-score by 3.1%.

TABLE I
QUANTITATIVE COMPARISON RESULTS OF ADHR-CDNET AND OTHER BENCHMARK METHODS ON THE LEBEDEV DATASET (THE BEST PERFORMANCE IS HIGHLIGHTED IN RED AND THE SECOND BEST IS HIGHLIGHTED IN BLUE)

| Method | Precision(%) | Recall(%) | F1-score(%) |
|---|---|---|---|
| CDNet[15] | 74.0 | 68.0 | 68.8 |
| FC-EF[16] | 81.6 | 76.1 | 78.8 |
| FC-Siam-conc[16] | 84.4 | 82.5 | 83.4 |
| FC-Siam-diff[16] | 85.8 | 83.6 | 84.5 |
| DTCDSCN[17] | 93.5 | 92.8 | 93.1 |
| DASNet[20] | 95.1 | 93.2 | 94.1 |
| STANet[21] | 96.4 | 91.4 | 93.8 |
| Si-HRNet[24] | 92.3 | 91.3 | 91.8 |
| TRCD[25] | 94.6 | 92.9 | 93.8 |
| DSAMNet[13] | 94.5 | 92.8 | 93.7 |
| DifUNet++[23] | 90.4 | 95.7 | 93.0 |
| DifUNet++DU[23] | 92.2 | 94.6 | 93.4 |
| ADHR-CDNet | 99.5 | 95.0 | 97.2 |

*b) Qualitative evaluation:* Fig. 3 shows some visualization comparison results of the proposed method and other comparison methods on the Lebedev dataset. From the first scene, it can be seen that although the shapes of the area marked by red boxes in (c1) are complex, our detection results shown in (h1) are very close to the ground truth. In the second scene, it can be seen that there are two red boxes in (c2), with a small vehicle target adjacent to the large building in the left box and multiple small vehicle targets in the right red box. Our method can accurately detect all the small objects as shown in (h2), but other methods such as FC-Siam-diff, STANet, and DASNet have false alarms or miss detection. In the third scene, the edges of the changed area in our result are more precise compared with the other methods. In the fourth scene, the roads which are not particularly visible in the red box can also be detected well by our method, as shown in (h4). In the fifth scene, even though the target road in (c5) is intricate, our method can still detect it well.

In conclusion, on the Lebedev dataset, our method can detect both large-scale and small-scale changed objects while still preserving the details of the changed area. It is possible because of our backbone's multiscale differential feature representation capability and MSSAM's effective multiscale feature fusion.

*2) On the LEVIR-CD Dataset:*

*a) Quantitative evaluation:* The quantitative comparison results on the LEVIR-CD dataset are shown in Table II. Our method performs best with the highest F1-score (91.4%). ADHR-CDNet increases the value of F1-score by 1.6%, compared with the second best one (ADS-Net).

*b) Qualitative evaluation:* Some visualization comparison results on the LEVIR-CD dataset are shown in Fig. 4. The outline shapes, edge locations, and area of the detected changed objects in our results are more accurate than those produced by other methods, especially the area marked by red boxes. Therefore, on the LEVIR-CD dataset, our method can also retain the detail information of the changed area.

*3) On the WHU Building Dataset:* Table III shows the quantitative comparison results with the SOTA methods on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

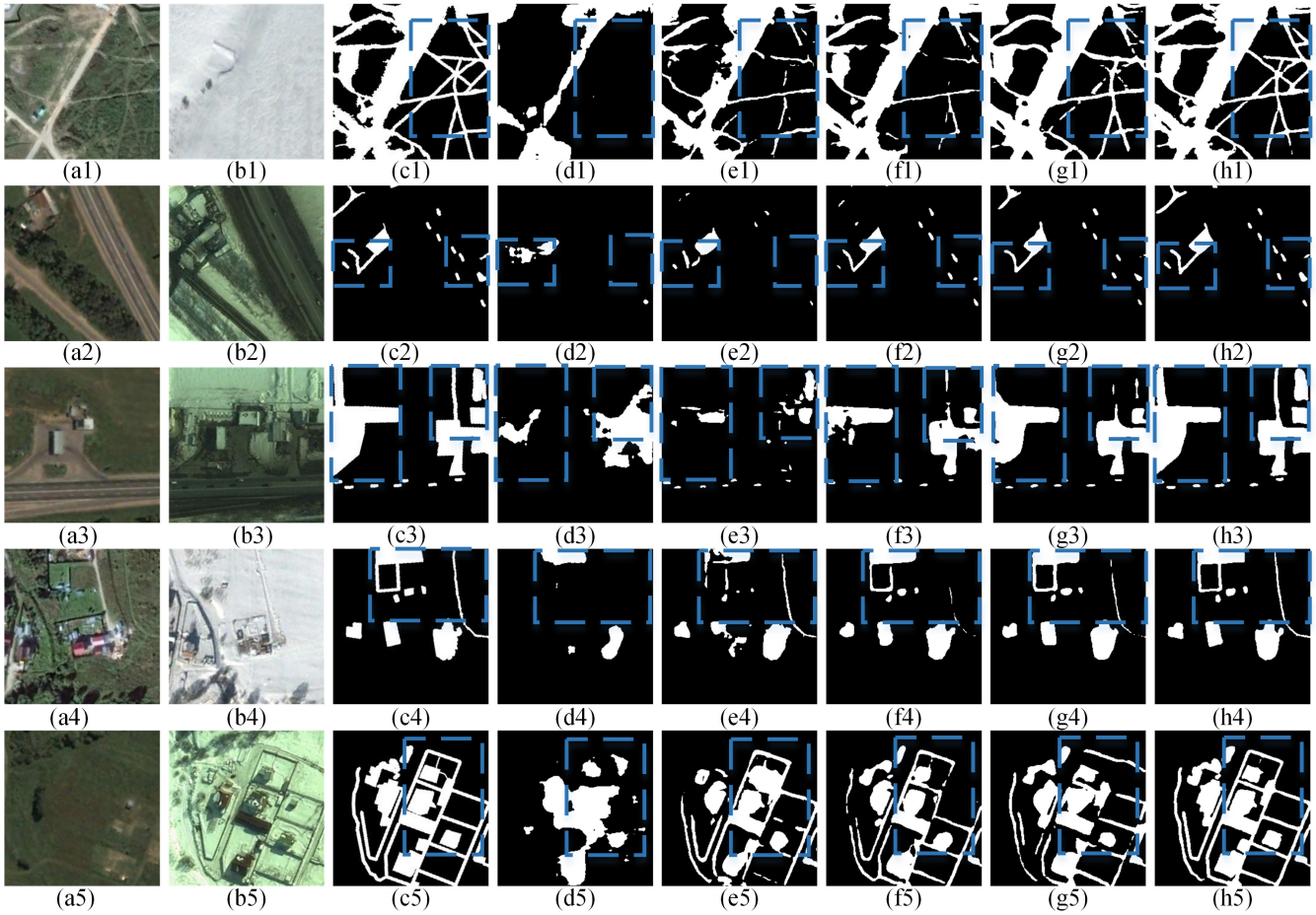



Fig. 3. Visualization comparison between ADHR-CDNet and other SOTA algorithms on the Lebedev dataset. (a) T1 image. (b) T2 image. (c) Ground truth. The predicted change maps of comparison networks are shown in (d) FC-Siam-diff. (e) STANet. (f) DASNet. (g) DifUNet++DU. (h) ADHR-CDNet.

TABLE II
QUANTITATIVE COMPARISON RESULTS OF ADHR-CDNET AND OTHER BENCHMARK METHODS ON THE LEVIR-CD DATASET (THE BEST PERFORMANCE IS HIGHLIGHTED IN RED AND THE SECOND BEST IS HIGHLIGHTED IN BLUE)

| Method | Precision(%) | Recall(%) | F1-score(%) |
|---|---|---|---|
| CDNet[15] | 82.9 | 88.9 | 85.8 |
| FC-EF[16] | 78.1 | 70.2 | 73.9 |
| FC-Siam-conc[16] | 79.2 | 78.0 | 78.6 |
| FC-Siam-diff[16] | 81.3 | 83.1 | 82.2 |
| DTCDSCN[17] | 86.5 | 89.2 | 87.8 |
| DASNet[20] | 85.3 | 86.1 | 85.7 |
| STANet[21] | 84.4 | 90.2 | 87.2 |
| EGRCNN[14] | 87.8 | 91.5 | 89.6 |
| TRCD [25] | 90.7 | 88.2 | 89.4 |
| DifUNet++[23] | 91.5 | 85.8 | 88.5 |
| DifUNet++DU[23] | 92.4 | 87.1 | 89.6 |
| DSA-Net[18] | 89.7 | 88.1 | 88.9 |
| Bit[29] | 89.2 | 89.4 | 89.3 |
| ADS-Net[19] | 89.7 | 91.4 | 89.8 |
| ADHR-CDNet | 94.6 | 88.5 | 91.4 |

WHU Building dataset. ADHR-CDNet achieves the best performance with the F1-score of 90.9%. It has outperformed all other networks. Compared with the second-best one (DTCDSCN), ADHR-CDNet increases the accuracy of F1-score by 1.2%. This verifies the superiority of our method in change detection.

### F. Parameter Tuning and Module Analysis

*1) Ablation Study:* To verify the effectiveness of MHRNet, DPM, and MSSAM, we conduct a series of ablation experiments on three experimental datasets. In detail, we separately add the module DPM and MSSAM into our backbone MHRNet to form two networks, named as MHRNet + DPM and MHRNet + MSSAM. Tables IV–VI show the quantitative results of ablation experiments on the Lebedev, LEVIR-CD, and WHU Building datasets, respectively.

Compared with the original high-resolution network, the F1-score of MHRNet is improved by 1.7%, 1.3%, and 1.1% on the Lebedev, LEVIR-CD, and WHU Building datasets, respectively.

By adding DPM to MHRNet, the value of F1-score increases by 1.6%, 0.9%, and 0.5% on the Lebedev, LEVIR-CD, and WHU Building datasets, respectively. The performance improvement is significant, since the differential feature information is particularly helpful for change detection to distinguish between substantive changes and pseudochanges. By adding MSSAM to MHRNet, the value of F1-score increases by 0.7%, 1.1%, and 0.8%, respectively. It verifies that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

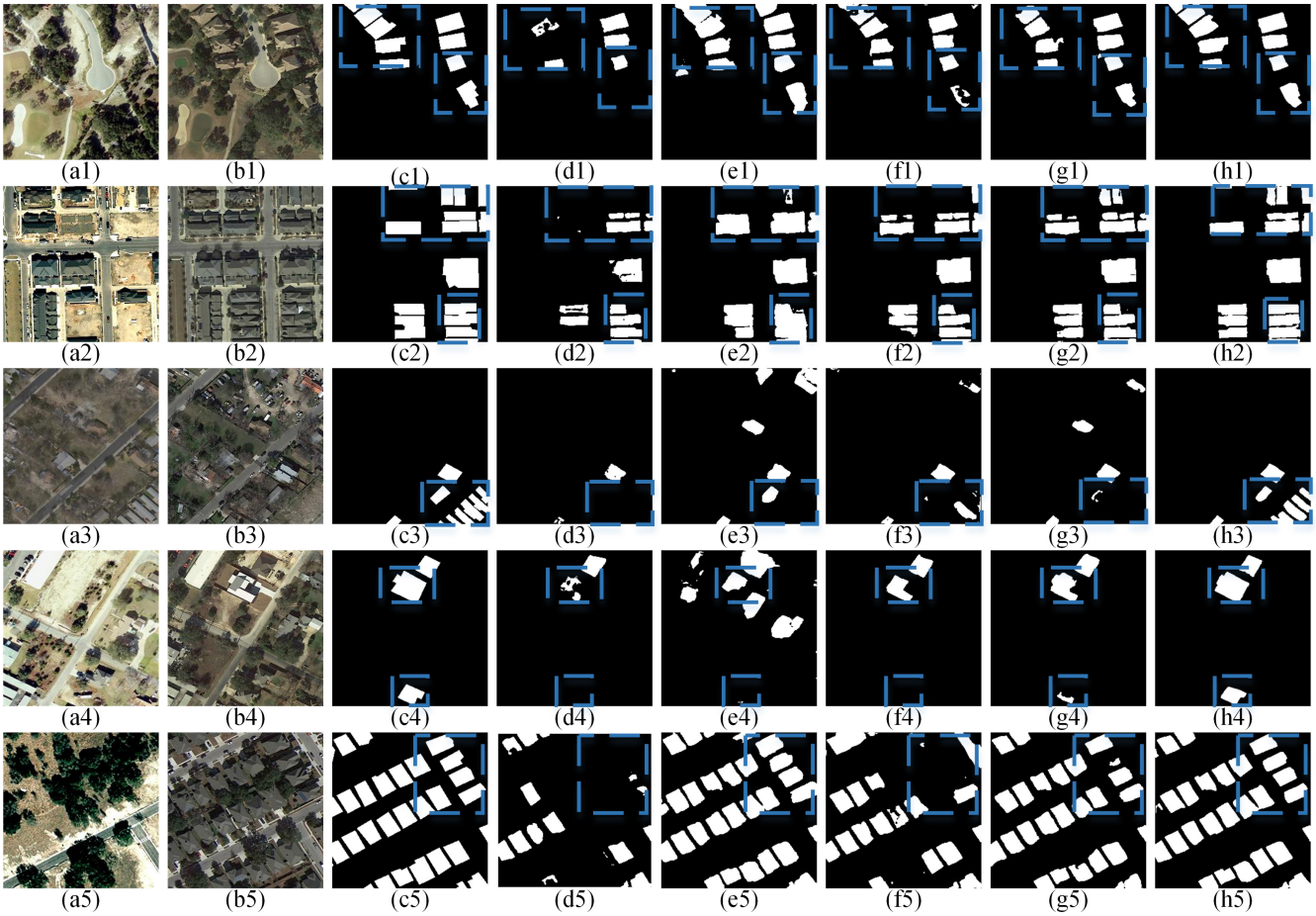



Fig. 4. Visualization comparison between ADHR-CDNet and other SOTA algorithms on the LEVIR-CD dataset. (a) T1 image. (b) T2 image. (c) Ground truth. The predicted change maps of comparison networks are shown in (d) FC-Siam-diff. (e) STANet. (f) DASNet. (g) DifUNet++DU. (h) ADHR-CDNet.

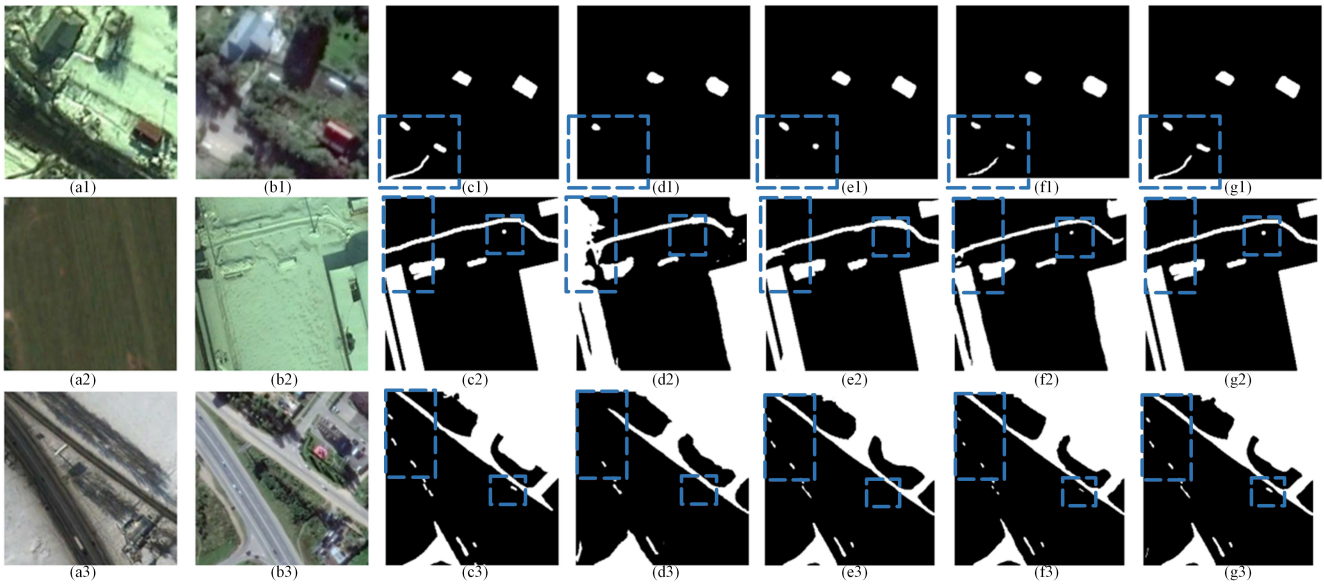



Fig. 5. Visualization results of ablation experiment on the Lebedev dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) MHRNet. (e) MHRNet + MSSAM. (f) MHRNet + DPM. (g) ADHR-CDNet.

our multiscale feature fusion strategy is effective. By adding both of them to MHRNet, the value of F1-score increases by 2.7%, 1.4%, and 1.3%, respectively. In addition, the F1-score of ADHR-CDNet(S) is 1.0%, 0.5%, and 0.4% lower than that of ADHR-CDNet on the Lebedev, LEVIR-CD, and WHU Building datasets, respectively. The lightweight model has a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE COMPARISON RESULTS OF ADHR-CDNET AND OTHER BENCHMARK METHODS ON THE WHU BUILDING DATASET (THE BEST PERFORMANCE IS HIGHLIGHTED IN RED AND THE SECOND BEST IS HIGHLIGHTED IN BLUE)

| Method | Precision(%) | Recall(%) | F1-score(%) |
|---|---|---|---|
| CDNet[15] | 82.8 | 85.8 | 84.3 |
| FC-EF[16] | 82.7 | 82.0 | 82.4 |
| FC-Siam-conc[16] | 73.7 | 74.5 | 74.1 |
| FC-Siam-diff[16] | 89.0 | 84.7 | 86.8 |
| DTCDSCN[17] | 90.2 | 89.4 | 89.7 |
| DASNet[20] | 85.1 | 85.7 | 85.4 |
| STANet[21] | 79.4 | 85.5 | 84.6 |
| EGRCNN[14] | 89.9 | 88.0 | 88.9 |
| Si-HRNet[24] | 94.0 | 79.4 | 86.1 |
| DifUNet++[23] | 88.6 | 84.7 | 86.6 |
| DifUNet++DU[23] | 89.0 | 85.4 | 87.2 |
| DSA-Net[18] | 89.4 | 87.6 | 88.5 |
| Bit[29] | 86.6 | 81.5 | 84.0 |
| ADHR-CDNet | 92.4 | 89.5 | 90.9 |

TABLE IV

ABLATION EXPERIMENT ON THE LEBEDEV DATASET

| Method | Precision(%) | Recall(%) | F1-score(%) |
|---|---|---|---|
| HRNet | 94.4 | 91.9 | 92.8 |
| MHRNet | 95.6 | 93.5 | 94.5 |
| MHRNet+DPM | 98.2 | 94.1 | 96.1 |
| MHRNet+MSSAM | 96.5 | 93.9 | 95.2 |
| ADHR-CDNet | **99.5** | **95.0** | **97.2** |
| ADHR-CDNet(S) | 98.1 | 94.3 | 96.2 |

TABLE V

ABLATION EXPERIMENT ON THE LEVIR-CD DATASET

| Method | Precision(%) | Recall(%) | F1-score(%) |
|---|---|---|---|
| HRNet | 91.2 | 86.3 | 88.7 |
| MHRNet | 92.9 | 87.2 | 90.0 |
| MHRNet+DPM | 93.8 | 88.1 | 90.9 |
| MHRNet+MSSAM | 94.0 | 88.3 | 91.1 |
| ADHR-CDNet | **94.6** | **88.5** | **91.4** |
| ADHR-CDNet(S) | 93.6 | 88.3 | 90.9 |

TABLE VI

ABLATION EXPERIMENT ON THE WHU BUILDING DATASET

| Method | Precision(%) | Recall(%) | F1-score(%) |
|---|---|---|---|
| HRNet | 89.2 | 87.8 | 88.5 |
| MHRNet | 90.3 | 88.9 | 89.6 |
| MHRNet+DPM | 91.2 | 89.1 | 90.1 |
| MHRNet+MSSAM | 91.6 | 89.2 | 90.4 |
| ADHR-CDNet | **92.4** | **89.5** | **90.9** |
| ADHR-CDNet(S) | 91.8 | 89.2 | 90.5 |

TABLE VII

ATTENTION ABLATION EXPERIMENT ON THE LEBEDEV DATASET

| Method | Precision(%) | Recall(%) | F1-score(%) |
|---|---|---|---|
| PAM[20] | 94.6 | 92.8 | 93.7 |
| CAM[20] | 91.8 | 90.6 | 91.2 |
| PAM+CAM[20] | 96.8 | 94.0 | 95.4 |
| BAM[21] | 96.2 | 93.1 | 94.6 |
| RA[62] | 96.5 | 93.6 | 95.0 |
| ADHR-CDNet | **99.5** | **95.0** | **97.2** |

TABLE VIII

DIFFERENCE OPERATORS EXPERIMENT ON THE LEBEDEV DATASET

| Method | Precision(%) | Recall(%) | F1-score(%) |
|---|---|---|---|
| NR[63] | 92.5 | 89.7 | 91.2 |
| GR[64] | 93.3 | 93.7 | 93.5 |
| ADHR-CDNet | **99.5** | **95.0** | **97.2** |

lower F1-score than the original model, but it is still superior to other current SOTA methods; see Tables IV–VI.

Table VII shows the comparison experiments between MSSAM and some other common attentions, such as position attention (PAM) [20], channel attention (CAM) [20], basic spatial–temporal attention module (BAM) [21], and RA [62]. We can see that MSSAM has better effect than other common attention and is more suitable for change detection of high-resolution remote sensing images.

The visualization results of the ablation experiment are shown in Fig. 5. Comparing the results of MHRNet and MHRNet + MSSAM, we can observe that the details of the detected changed areas of MHRNet + MSSAM are more accurate, especially in the area represented by red boxes. Moreover, MHRNet + MSSAM can detect more minor changes. It can demonstrate that MSSAM is reasonable for multiscale feature fusion. Comparing the results of MHRNet and MHRNet + DPM, MHRNet + DPM can detect more changed objects, especially the unobvious changed objects. The reason for that is DPM can improve the ability to encode the unobvious changed area by extracting multiscale differential information. AtDifHRNet combines the advantages of DPM and MSSAM, resulting in a significantly better and more accurate detection result.

*2) Experiments on Difference Operators:* In DPM, absolute difference is applied to obtain the difference information. To verify the effect of different difference operators, we replace the absolute difference operation between the feature map pairs of different scales in DPM with neighborhood-based ratio (NR) [63] and Gauss-log ratio (GR) [64], respectively, which have been adopted in change detection. The comparison results are shown in Table VIII. It can be seen that absolute difference performs much better than other two difference operators for DPM.

*3) Experiments on Convolution Kernel Number in MSSAM:* To verify the influence of the convolution kernel number in MSSAM on detection performance, we conduct a group of experiment, as shown in Fig. 6. The horizontal axis represents the number of different filters in MSSAM, and the vertical axis represents the corresponding F1-score. The experimental results show that our network achieves the best performance when the number of convolution kernel is 512.

When the number of convolution kernels increases from 128 to 256 and 512, the detection performance will gradually increase. However, when the number of convolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

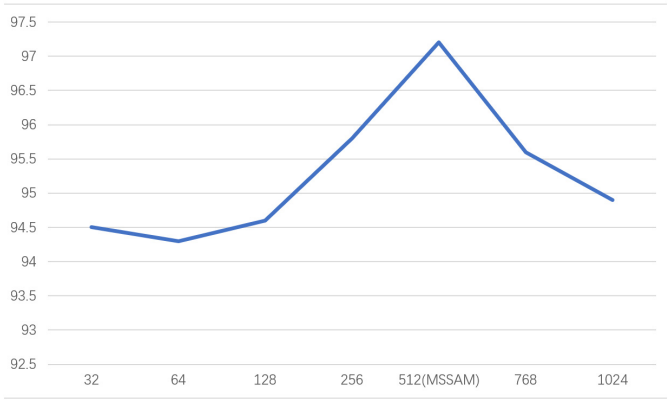



Fig. 6. Influence of convolution kernel number in MSSAM.

TABLE IX
TRAINING EFFICIENCY EXPERIMENT ON THE LEBEDEV DATASET (BATCHSIZE = 2, AN ITERATION)

| Method | Time ($10^{-2}$s) | F1-score(%) |
|---|---|---|
| FC-EF[16] | 11 | 78.8 |
| FC-Siam-conc[16] | 14 | 83.4 |
| FC-Siam-diff[16] | 12 | 84.5 |
| DifUNet++DU[23] | 18 | 93.4 |
| DTCDSCN[17] | 27 | 93.1 |
| STANet[21] | 16 | 93.8 |
| DASNet[20] | 18 | 94.1 |
| ADHR-CDNet(S) | 17 | 96.2 |
| ADHR-CDNet | 29 | 97.2 |

kernels is greater than 512, the performance decreases. The possible reason is that more convolution kernels will make the module MSSAM own more powerful representing ability, leading to better results, but too much may cause overfitting.

### G. Training Efficiency Comparison

To compare the training efficiency of different models, we conduct a comparison experiment on the Lebedev dataset, as shown in Table IX. For all the models, the same experimental hyperparameters are adopted to fairly evaluate the training efficiency, and we record the training time spent on an iteration for each compared model.

It can be seen from Table IX, the most competitive model is DASNet. Compared with DASNet, the F1-score of our model ADHR-CDNet(S) is 2.1% higher and the training time (batchsize = 2, an iteration) is 0.01 s faster, that is, the speed is increased by 5.6%. Besides, using ADHR-CDNet can further improve the F1-score to 97.2% with not too much time overhead.

## V. CONCLUSION

In this article, an effective remote sensing image change detection network named ADHR-CDNet is proposed. A high-resolution backbone based on HRNet is designed to obtain multilevel and multiscale features. The DPM is then adopted to extract multiscale differential features from two input images, which are integrated into our backbone as supplementary input. DPM significantly improves the accuracy of change detection, since it is helpful to distinguish between substantive changes and pseudochanges caused by interference factors. Furthermore, MSSAM is presented to learn the weight maps for different scale feature groups. Finally, MSSAM is used to fuse the spatial detail information of multiscale feature groups produced by our backbone to generate finer prediction. The quantitative and qualitative experimental results demonstrate that our method performs much better than the SOTA methods.

Most change detection methods including our method only judge whether an area is changed, but cannot predict which classification of the changed area is. In the future, we will focus on semantic change detection, which can predict not only whether an area is changed but also the classification of the changed area.

## REFERENCES

[1] J. Chen, M. Lu, X. Chen, J. Chen, and L. Chen, "A spectral gradient difference based approach for land cover change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 85, pp. 1–12, Nov. 2013.
[2] J. Verbesselt, R. Hyndman, G. Newnham, and D. Culvenor, "Detecting trend and seasonal changes in satellite image time series," *Remote Sens. Environ.*, vol. 114, no. 1, pp. 106–115, 2010.
[3] E. Hamunyela, J. Verbesselt, and M. Herold, "Using spatial context to improve early detection of deforestation from landsat time series," *Remote Sens. Environ.*, vol. 172, pp. 126–138, Jan. 2016.
[4] D. Brunner, G. Lemoine, and L. Bruzzone, "Earthquake damage assessment of buildings using VHR optical and SAR imagery," *IEEE Trans. Geosc. Remote Sens.*, vol. 48, no. 5, pp. 2403–2420, May 2010.
[5] N. Longbotham et al., "Multi-modal change detection, application to the detection of flooded areas: Outcome of the 2009–2010 data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 1, pp. 331–342, Feb. 2012.
[6] G. Xian and C. Homer, "Updating the 2001 national land cover database impervious surface products to 2006 using Landsat imagery change detection methods," *Remote Sens. Environ.*, vol. 114, no. 8, pp. 1676–1686, Aug. 2010.
[7] P. Zhang, M. Gong, L. Su, J. Liu, and Z. Li, "Change detection based on deep feature representation and mapping transformation for multi-spatial-resolution remote sensing images," *Photogram. Remote Sens.*, vol. 116, pp. 24–41, Sep. 2016.
[8] C. Zhang, S. Wei, S. Ji, and M. Lu, "Detecting large-scale urban land cover changes from very high resolution remote sensing images using CNN-based classification," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 4, p. 189, Apr. 2019.
[9] T. Liu, Y. Li, Y. Cao, and Q. Shen, "Change detection in multitemporal synthetic aperture radar images using dual-channel convolutional neural network," *J. Appl. Remote Sens.*, vol. 11, no. 4, 2017, Art. no. 042615.
[10] E. Guo et al., "Learning to measure change: Fully convolutional Siamese metric networks for scene change detection," 2018, *arXiv:1810.09111*.
[11] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.
[12] M. Zhang, G. Xu, K. Chen, M. Yan, and X. Sun, "Triplet-based semantic relation learning for aerial remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 266–270, Feb. 2019.
[13] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.
[14] B. Bai, W. Fu, T. Lu, and S. Li, "Edge-guided recurrent convolutional neural network for multitemporal remote sensing image building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.
[15] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.
[16] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

[17] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[18] Q. Ding, Z. Shao, X. Huang, and O. Altan, "DSA-Net: A novel deeply supervised attention-guided network for building change detection in high-resolution remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102591.

[19] D. Wang, X. Chen, M. Jiang, S. Du, B. Xu, and J. Wang, "ADS-Net: An attention-based deeply supervised network for remote sensing image change detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 101, Sep. 2021, Art. no. 102348.

[20] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[21] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[22] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, 2019.

[23] X. Zhang et al., "DifUnet++: A satellite images change detection network based on Unet++ and differential pyramid," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3049370.

[24] Z. Cao, M. Wu, R. Yan, F. Zhang, and X. Wan, "Detection of small changed regions in remote sensing imagery using convolutional neural network," *IOP Conf. Earth Environ. Sci.*, vol. 502, no. 1, 2020, Art. no. 012017.

[25] F. Pan, Z. Wu, X. Jia, Q. Liu, Y. Xu, and Z. Wei, "A temporal-reliable method for change detection in high-resolution bi-temporal remote sensing images," *Remote Sens.*, vol. 14, no. 13, p. 3100, Jun. 2022.

[26] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[27] T. Lei et al., "Difference enhancement and Spatial–Spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[28] H. Zhang, M. Gong, P. Zhang, L. Su, and J. Shi, "Feature-level change detection using deep representation and feature change analysis for multispectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 11, pp. 1666–1670, Nov. 2016.

[29] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," 2021, *arXiv:2103.00208*.

[30] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2016.

[31] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Jul. 2020.

[32] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, pp. 565–571, May 2018, doi: 10.5194/isprs-archives-XLII-2-565-2018.

[33] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[34] C. Wu, H. Chen, B. Do, and L. Zhang, "Unsupervised change detection in multi-temporal VHR images based on deep kernel PCA convolutional mapping network," 2019, *arXiv:1912.08628*.

[35] X. Qu, F. Gao, J. Dong, Q. Du, and H.-C. Li, "Change detection in synthetic aperture radar images using a dual-domain network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3073900.

[36] L. T. Luppino et al., "Deep image translation with an affinity-based change prior for unsupervised multimodal change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–22, 2022, doi: 10.1109/TGRS.2021.3056196.

[37] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.

[38] Y. Gao, F. Gao, J. Dong, and S. Wang, "Change detection from synthetic aperture radar images based on channel weighting-based deep cascade network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 11, pp. 4517–4529, Nov. 2019.

[39] J. Geng, H. Wang, J. Fan, and X. Ma, "Change detection of SAR images based on supervised contractive autoencoders and fuzzy clustering," in *Proc. Int. Workshop Remote Sens. Intell. Process. (RSIP)*, May 2017, pp. 1–3.

[40] L. Su, M. Gong, P. Zhang, M. Zhang, J. Liu, and H. Yang, "Deep learning and mapping based ternary change detection for information unbalanced images," *Pattern Recognit.*, vol. 66, pp. 213–228, Jun. 2017.

[41] X. Niu, M. Gong, T. Zhan, and Y. Yang, "A conditional adversarial network for change detection in heterogeneous images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 45–49, Jan. 2018.

[42] M. Gong, T. Zhan, P. Zhang, and Q. Miao, "Superpixel-based difference representation learning for change detection in multispectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2658–2673, May 2017.

[43] H. Fang, P. Du, and X. Wang, "A novel unsupervised binary change detection method for VHR optical remote sensing imagery over urban areas," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 108, Apr. 2022, Art. no. 102749.

[44] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Aug. 2020, doi: 10.1109/TGRS.2020.3011913.

[45] H. Chen, W. Li, and Z. Shi, "Adversarial instance augmentation for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, doi: 10.1109/TGRS.2021.3066802.

[46] B. Hou, Q. Liu, H. Wang, and Y. Wang, "From W-Net to CDGAN: Bitemporal change detection via deep learning techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1790–1802, Mar. 2020.

[47] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[48] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[49] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[50] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[51] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[52] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[53] W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Asymmetric feature fusion network for hyperspectral and SAR image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 18, 2022, doi: 10.1109/TNNLS.2022.3149394.

[54] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021.

[55] D. Toth, T. Aach, and V. Metzler, "Illumination-invariant change detection," in *Proc. 4th IEEE Southwest Symp. Image Anal. Interpretation*, 2000, pp. 3–7.

[56] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[57] L.-C. Chen, Y. Yang, J. Wang, W. Xu, and A. L. Yuille, "Attention to scale: Scale-aware semantic image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3640–3649.

[58] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Cham, Switzerland: Springer, 2015, pp. 234–241.

[59] Z. Zhou et al., "UNet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Dec. 2019.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.

[60] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[61] F. Wang et al., "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3156–3164.
[62] K. Zhu and J. Wu, "Residual attention: A simple but effective method for multi-label recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 184–193.
[63] M. Gong, Y. Cao, and Q. Wu, "A neighborhood-based ratio approach for change detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 2, pp. 307–311, Mar. 2012.
[64] B. Hou, Q. Wei, Y. Zheng, and S. Wang, "Unsupervised change detection in SAR image based on gauss-log ratio image fusion and compressed projection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 8, pp. 3297–3317, Aug. 2014.



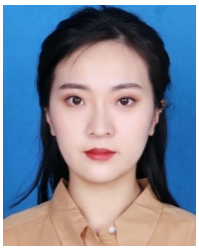
**Xiuwei Zhang** received the B.S., M.S., and Ph.D. degrees from the School of Computer Science, Northwestern Polytechnical University, Xi'an, China, in 2004, 2007, and 2011, respectively.

She is now an Associate Professor with the School of Computer Science, Northwestern Polytechnical University. Her research interests include remote sensing image processing, multimodel image fusion, image registration, and intelligent forecasting.



**Mu Tian** received the bachelor's degree from the School of Computer Science, Northwestern Polytechnical University, Xi'an, China, in 2019, where he is currently pursuing the master's degree in computer technology.

His research interests include machine learning, deep learning, computer vision, and change detection.



**Yinghui Xing** (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2014 and 2020, respectively.

She is now an Associate Professor with the School of Computer Science, Northwestern Polytechnical University, Xi'an. Her research interests include remote sensing image processing, image fusion, and image super-resolution.



**Yuanzeng Yue** received the bachelor's degree in software engineering from Northwestern Polytechnical University, Xi'an, China, in 2021.

His research interests include change detection and image super-resolution.



**Yanping Li** received the bachelor's degree in computer science and technology from the Shaanxi University of Technology, Hanzhong, China, in 2022. She is currently pursuing the master's degree in electronic information with Northwestern Polytechnical University, Xi'an, China.

Her research interests include machine learning, deep learning, computer vision, and remote sensing image processing.



**Hanlin Yin** received the B.S. degree in control science and engineering from the Harbin Institute of Technology, Harbin, China, in 2009, and the Ph.D. degree in control science and engineering from Xi'an Jiaotong University, Xi'an, China, in 2016.

He is currently an Assistant Professor with the School of Computer Science, Northwestern Polytechnical University, Xi'an. His research interests include rainfall-runoff modeling, time series prediction, information fusion, and performance evaluation.



**Runliang Xia** received the B.S. and M.S. degrees from the School of Computer and Information Engineering, Hohai University, Nanjing, China, and the Ph.D. degree from the School of Water Resources and Hydropower Engineering, Wuhan University, Wuhan, China, in 2006, 2009, and 2013, respectively.

He is now a Professorate Senior Engineer with the Yellow River Institute of Hydraulic Research, Zhengzhou, China. His research interests include water resource informatization and numerical simulation of water conservancy.



**Jin Jin** received the bachelor's degree from North China University of Water Resources and Electric Power, Zhengzhou, China, in 2003, and is currently pursuing the Ph.D. degree in computer science at Northwestern Polytechnical University.

He is currently a Senior Engineer with the Yellow River Institute of Hydraulic Research, Zhengzhou, China. His main research interests are water conservancy project management and safety monitoring, artificial intelligence rainfall-runoff modeling, and hydro-meteorological big data application.



**Yanning Zhang** (Senior Member, IEEE) received the B.S. degree from the Dalian University of Science and Engineering, Dalian, China, in 1988, and the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 1993 and 1996, respectively.

She is currently a Professor with the School of Computer Science, Northwestern Polytechnical University. Her research interests include signal and image processing, computer vision, and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:46:46 UTC from IEEE Xplore. Restrictions apply.